

INTRODUCTION TO INFO. & COMM. TECHNOLOGY

20-09-2023

About Me

Shazia Rizwan

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- Office hours: Mon -Fri 10:00am - 4:00pm
(appointment strongly recommended)

General Rules

- You may expect a *Marked* activity in any class (un announced)
- No Makeup for an announced quiz and assignment
- Attendance will be marked in first **20mins** of class
- No updating possible once the attendance is entered in CMS
- Zero tolerance towards cheating and plagiarism

Course Rules

- No marked activity will be dropped
- Keep a track of your attendance yourself - 75% required to be eligible to sit in the Final Exam
- Feel free to see me or the course TA for extra help/tutoring
- will be there in the lab classes
- Keep an eye on Google classroom for any announcements or postings

Books - for reference

- Understanding Computers - Today and Tomorrow Comprehensive 2016 by Deborah Morley, Charles S. Parker
- Computer Fundamentals by Anita Goel
- Starting out with C++ Early Objects, 10th Edition, Tony Gaddis, Judy Walters, Godfrey Muganda
- **Softcopies available**



Don't miss class !

ASK QUESTIONS !

READ BOOK !

Work HARD

Division of Marks

Class Tests (3)	15
Lab Assignments (In Class + Labs)	15
Project	05
Mid Term Tests (2)	25
Final Exam	40
Total	<u>100</u>

Important

- Grading → Relative
- Google Classroom Code → 7mbmqbd

Course Objectives

- To understand the basics of computing technology
- To show how computer hardware represents information using binary number system.
- To learn process of algorithm development
- To build sets of test data to verify the correct functioning of the algorithm
- To be able to design and use sequence, selection and iteration constructs in algorithms.

What is Computer Science?

- It is a collection of theories and concepts to learn how to use computers to design and develop solutions for the problems surrounding you
- Foundation of computer science → problem solving process
- It is much more than coding..
- Cannot survive in CS without analytical thinking
- So..

Start thinking and analyzing to design innovative solutions

Why Study Computer Science?

- It allows you to program a computer to do what you want it to
- It is applicable in all disciplines today
- It drives innovation across all areas of study today
- It enables us to change the world around us through creativity and computational thinking
- You will learn how information is stored and processed to get meaningful results
- You will learn new technologies and skills which will make you highly employable

What is a Computer ?

- The word computer comes from the word "compute" → to calculate
- A computer is an electronic device that can perform arithmetic operations at high speed
- It can store, process and retrieve data where desired

Basic Operations

- **Input**—entering data into the computer
- **Processing**—performing operations on the data
- **Output**—presenting the results
- **Storage**—saving data, programs, or output for future use

Lets List down parts of a computer

- The physical parts are called **Computer Hardware**
- The tangible parts

Lets List down parts of a computer

- The physical parts are called **Computer Hardware**
- Keyboard
- Mouse
- Display Screen/ Monitor
- Speakers
- System Unit
- Microphone
- External storage Devices
- CD/DVD Drives in system unit/external
- Flash Memory
- Printer
- WebCam

Any More Tangible devices?

- Scanner (Flatbed/handheld)
- Barcode Reader
- Motherboard
- JoyStick
- Projector
- Headphones
- Random Access Memory (RAM) → temporarily stores data and serves as the computer's "working memory → volatile
- Hard Disk Drive (HDD) → OS + software + files
- Hard drives are non-volatile

Parts of a computer

- Solid State Drive (SSD) → almost identical to traditional hard disk drives from the outside, they have no moving parts. This means that they use less power, access data faster, and are generally more reliable
- External hard drive

Parts of a computer

- The Intangible parts

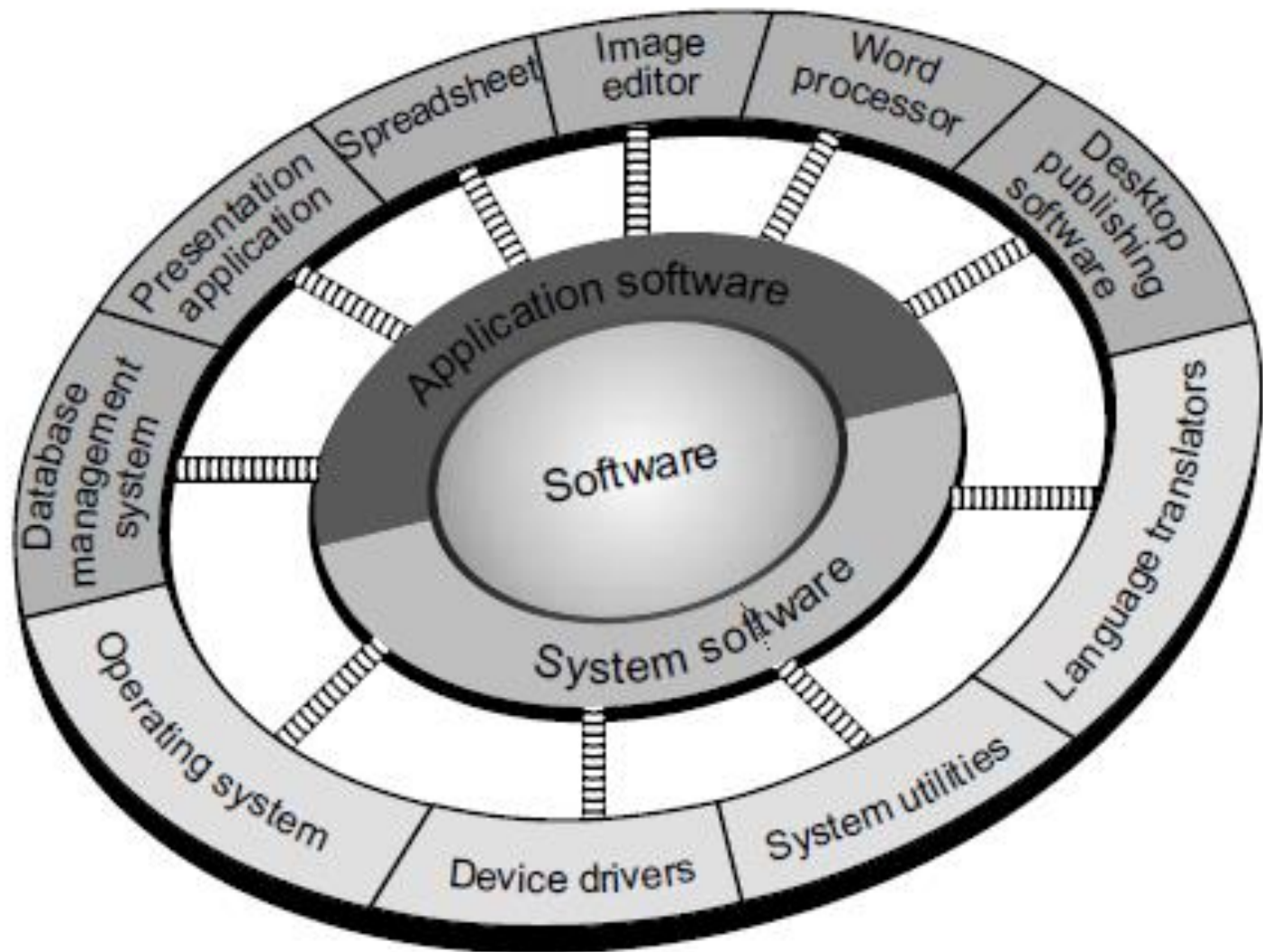
Let's List down

- The Intangible parts
- Programs
- Application software
- System software

Application SW

- Programs which allow people to perform specific tasks using a computer
- Creating letters
- Preparing budgets
- Managing inventory
- Playing games
- Watching videos
- Listening to music
- Editing digital photographs
- Designing homes
- Viewing Web pages
- Exchanging e-mails and much more..

SW Categories



One more component of a computer System

- Liveware
- AKA Humanware
- It is the computer user
- User commands the computer to execute an instruction

Classification of Devices

- Input Devices
- Output Devices
- Storage Devices
- Are these resources of a computing device?

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- Are these resources of a computing device?
- Yes
- What are the other resources?

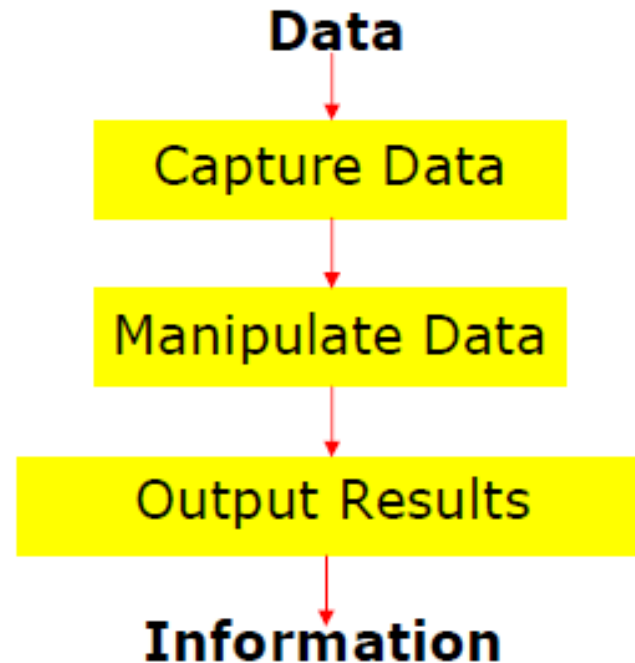
Classification of Devices

- Input Devices
- Output Devices
- Storage Devices

- Are these resources of a computing device?
- Yes
- What are the other resources?
- Software
- Memory
- Data
- CPU?

Data Vs Information

- a user inputs **data** into a computer
- Almost any kind of fact or set of facts can become computer data: text, numbers, images, notes in a song etc.
- When data is processed into a meaningful form, it becomes **information**



Information

- Information is frequently generated to answer some type of question:
- How many employees of BNU worked less than 40 hours per week in the month of September
- How many seats are available on a particular flight from Lahore to Karachi today
- How many candidates got selected but did not join BNU in Fall 2023
- Number of Male and female voters in Punjab for 2023

Information Processing

- *Information processing* → the conversion of data into information

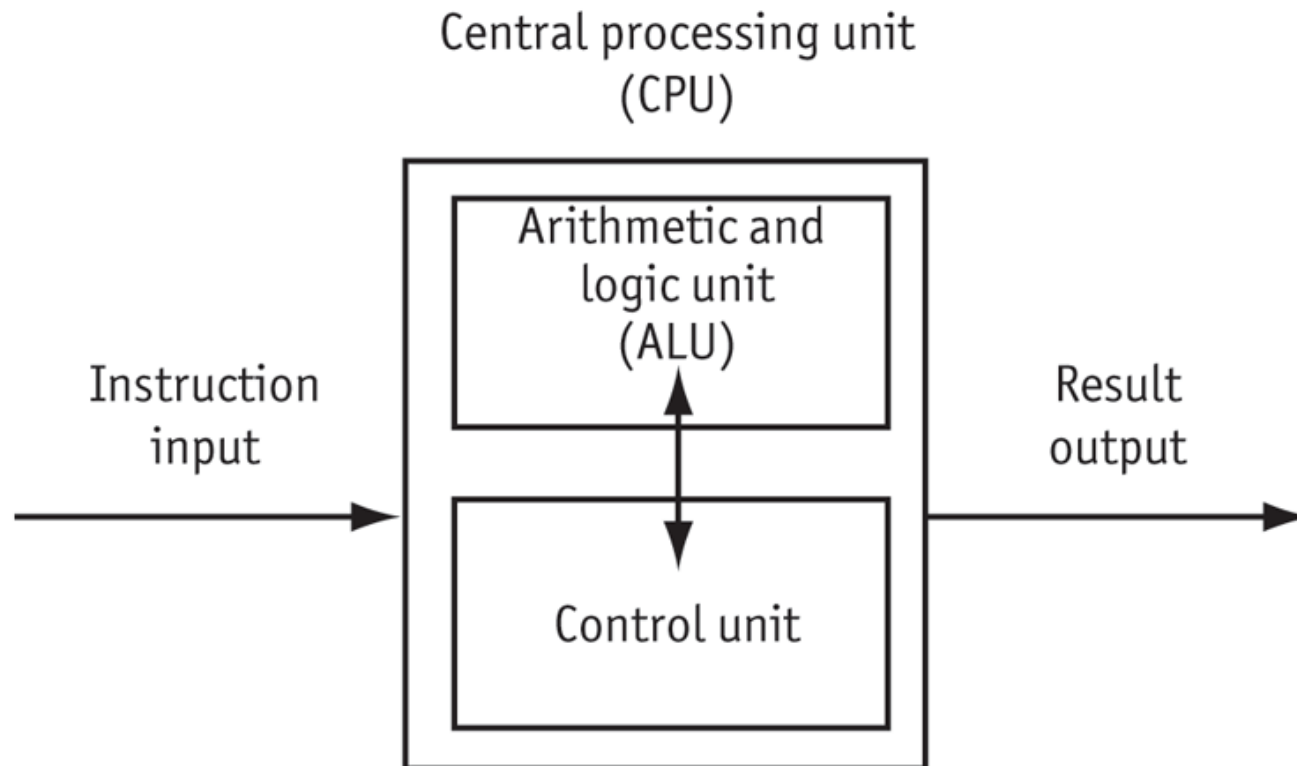
CPU : the main processing device

- The *central processing unit*, or *CPU* → *brain of a computer*
- CPUs, known as *microprocessors*, are tiny chips small enough to be held in the palm of your hand
- Located inside the system unit and performs the calculations and comparisons needed for processing; it also controls the computer's operations
- Various types of *memory* located inside the system unit are used to store data and instructions while the CPU is working
- Additional processors such as the *graphics processing unit* (*GPU*)

Different types of CPU

- Single Core
 - Dual Core
 - Quad Core
 - Hexa Core
 - Octa Core
 - Deca Core
-
- **RIY**

CPU - Structure



Internal Structure of CPU

- The central processing unit consists of two parts:
- The *control unit*
- The *arithmetic and logic unit (ALU)*

Control Unit

- To manage and coordinate the operations of the CPU
- Controls the flow of data and instructions between CPU and other parts of the computer system
- Interprets and executes instructions from computer memory
- Fetches instructions from main memory (RAM) and decodes them
- Determines the operation to be performed
- It directs the components within the CPU to carry out those operations

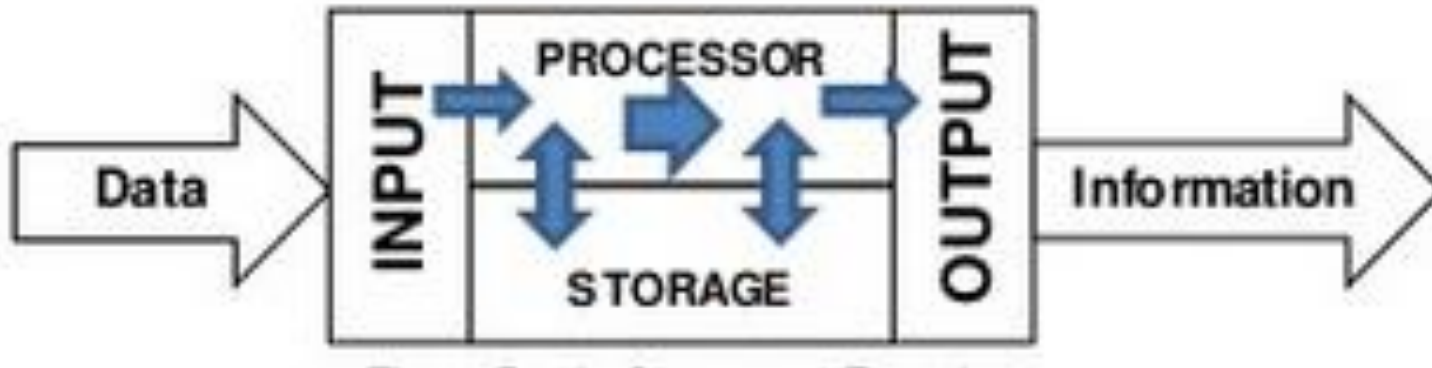
ALU

- Functions are:
- Arithmetic operations → addition, subtraction, division, multiplication
- Logical Operations → AND, OR, Equal, Less Than, Greater than

CPU

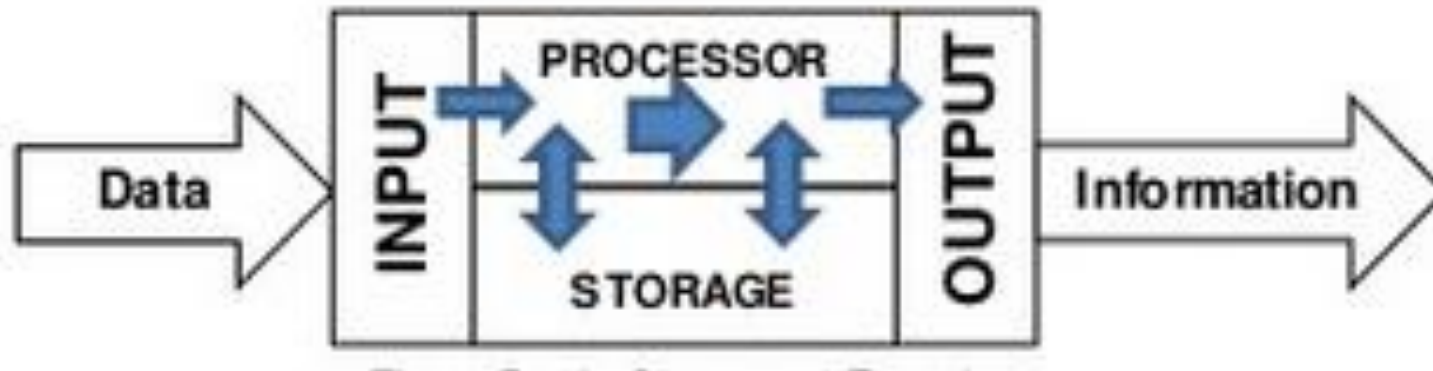
- CPU is engaged in a process known formally as the *fetch/decode/execute cycle*
- These steps are repeated as long as there are instructions to perform

Data Processing Cycle

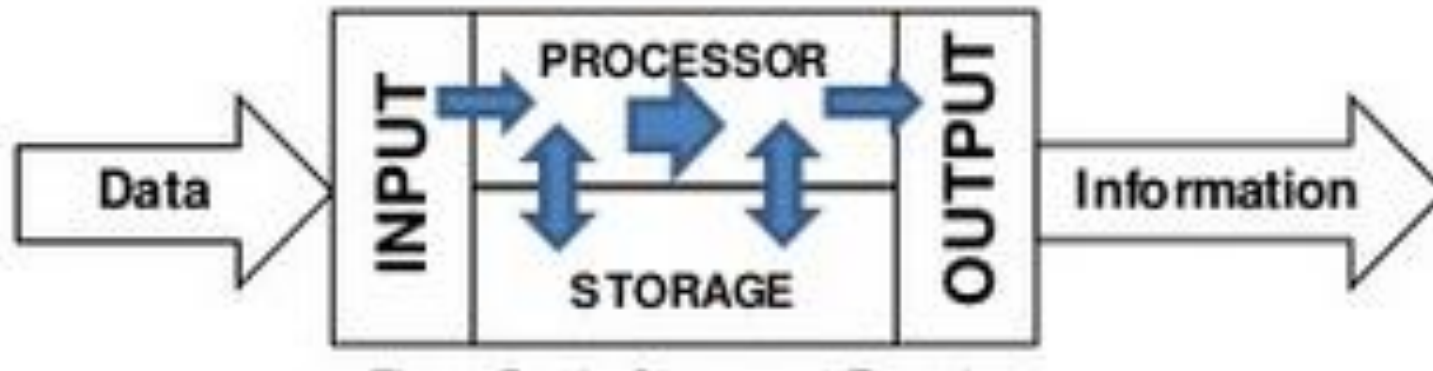


Data is a collection of organized facts

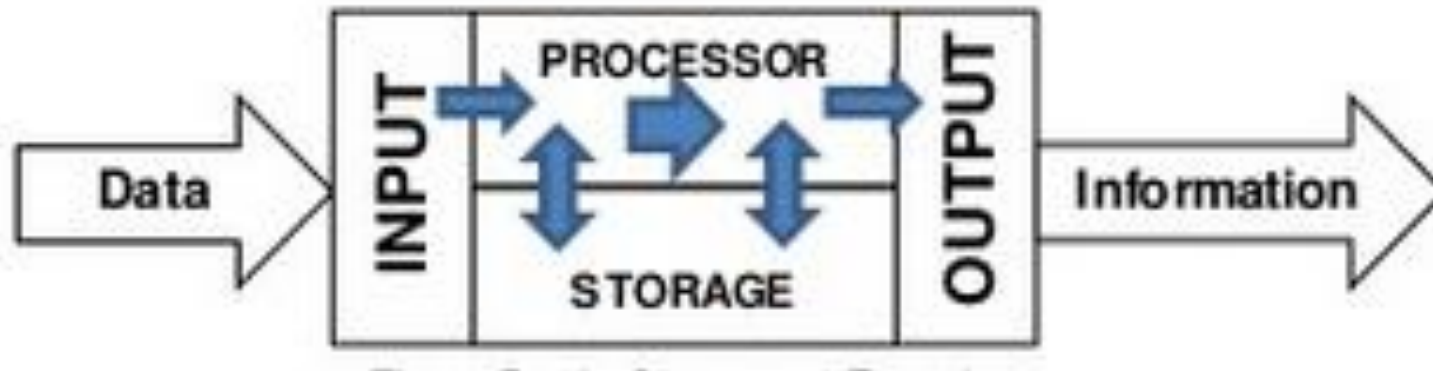
Information is processed data → Meaningful form of data



Input Unit : Accepts data

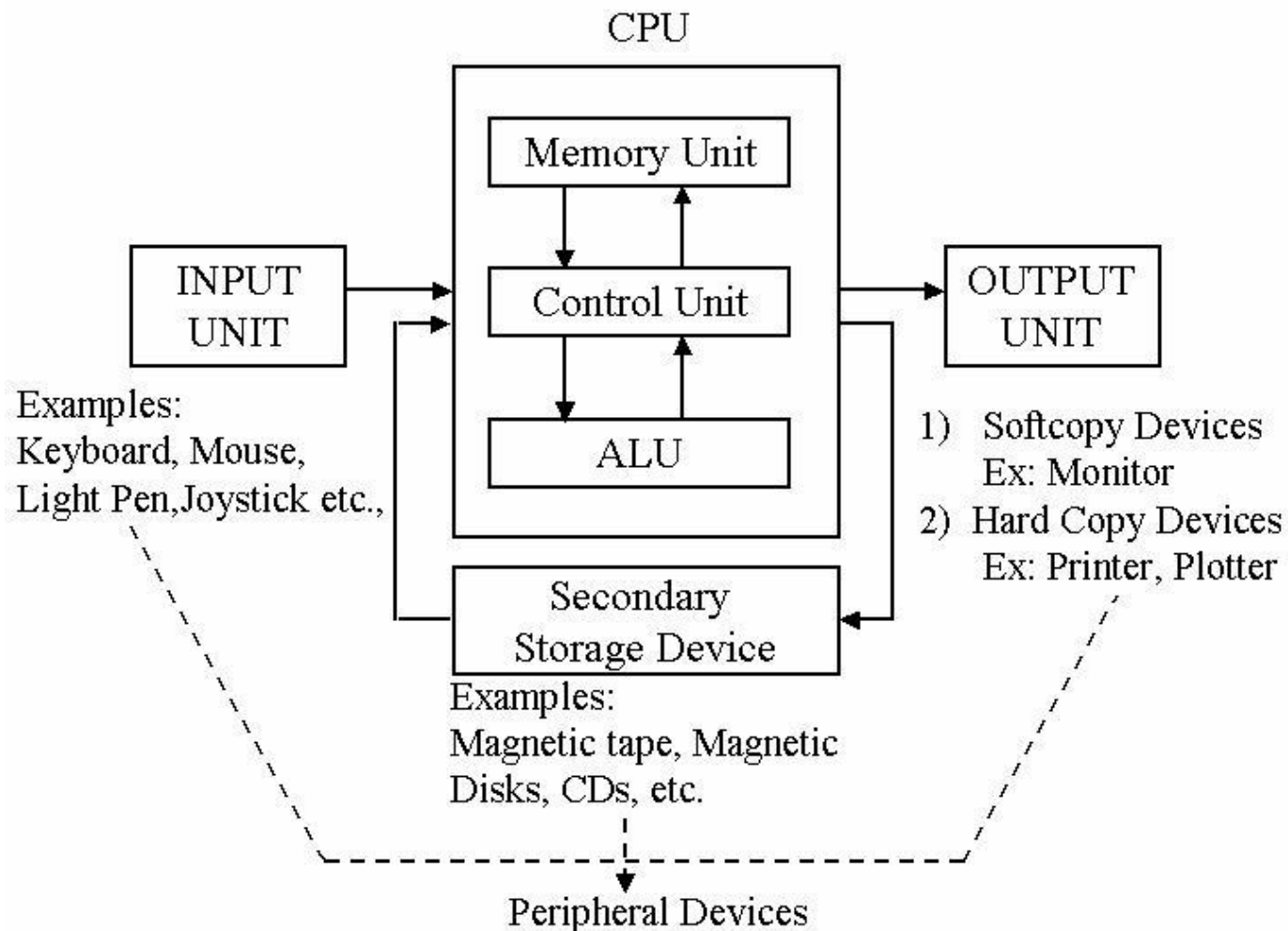


Processor : Processes data to convert it into information



Output Unit: Presents information to the user

Components of a computer internally



Data Processing Cycle

- Inputs
- Processing
- Outputs
- Storage (if required)

Computer Users

- All software is created by individuals known as programmers or software developers
- Highly trained and skilled people to design, create, and test computer programs
- Other *computer professionals* include *systems analysts* → design computer systems to be used
- *Computer operations personnel* → responsible for the day-to-day computer operations, such as maintaining systems or troubleshooting user-related problems
- *Security specialists* → responsible for securing the computers and networks against *hackers* and other intruders

Program

- What is needed for a well designed computer program?
- The logical flow of the instructions
- The mathematical procedures
- The appearance of the screens
- The way information is presented to the user
- The program's "user-friendliness"
- Documentation, help files, tutorial, etc.

Programming

- Translating algorithms into a formal Programming language
- Language that can be understood by both humans and computers
- Computers always convert the code into machine code before executing it

Is Programming same as Computer Science?

- Computer science is the study of what computers [can] do
- Programming is the practice of making computers do things

Computers to fit every need

- **Embedded computers** are built into products (such as cars and household appliances)
- To provide added functionality
- **Mobile devices** are small devices (such as *mobile phones* and **tablets**) with computing or Internet capabilities
- An Internet-enabled mobile phone is called a **smartphone**

Class Discussion

- List down at least 10 Input Devices

Class Discussion

- List down at least 10 Input Devices:
 - i. Keyboard
 - ii. Graphics tablet
 - iii. Barcode reader
 - iv. Gamepad
 - v. Joystick
 - vi. Microphone
 - vii. Mouse (pointing device)
 - viii. Scanner
 - ix. Webcam
 - x. Touchpad
 - xi. Light pen
 - xii. Microphone
 - xiii. Electronic whiteboard

Class Discussion

- List down at least 10 Output Devices

Class Discussion

- List down at least 10 Output Devices:
- Monitor
- Printer
- Plotter
- Projector
- Speakers
- Touch Screens (I/O)
- CD/DVD (I/O)
- Storage devices (I/O)

Let's write our first program

- Using simple English
- Problem: User enters two integers. Your program should give you two answers: Sum & Product

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- Problem: User enters two integers. Your program should give you two answers: Sum & Product
- **Inputs:**
 - Integer1
 - Integer2

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- Outputs:

Let's write our first program

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- Problem: User enters two integers. Your program should give you two answers: Sum & Product
- Outputs:
 - Sum
 - Product

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Sum = Integer1 + Integer2

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- Problem: User enters two integers. Your program should give you two answers: Sum & Product
- Required Processing:

Sum = Integer1 + Integer2

Product = Integer1 * Integer2

Let's write our first program

- Using simple English
- Problem: User enters two integers. Your program should give you two answers: Sum & Product
- **Required Processing:**
- $\text{Sum} = \text{Integer1} + \text{Integer2}$
- $\text{Product} = \text{Integer1} * \text{Integer2}$

- Assignment operator ?
- Addition symbol ?
- Multiplication symbol ?

Let's write our first program

- Using simple English
- Problem: User enters two integers. Your program should give you two answers: Sum & Product
- **Required Processing:**
- $\text{Sum} = \text{Integer1} + \text{Integer2}$
- $\text{Product} = \text{Integer1} * \text{Integer2}$

- Assignment operator $\rightarrow$ equal to sign
- Addition symbol $\rightarrow$ Plus sign
- Multiplication symbol $\rightarrow$ Asterisk sign

We will write one program in each class using English before we jump into C++ language

Ref Chapters

- Book A
- Chp 1